

4 (**short**) Water of density ρ is confined by a solid gate of length $4R$ under the effect of gravity g , as shown in Fig. 1. The problem is homogeneous in the direction into the page, and the gate can rotate about a pivot freely in the clockwise direction, but is prevented from rotating anti-clockwise beyond the vertical position by a backstop as shown in the figure. The section of the gate below the pivot is in a circular cavity of radius R that prevents water leakage. The water level is h .

(a) Find the value of h for which the gate will begin to open. [5]

(b) For h lower than the value found in (a), find the force F produced by the hydrostatic pressure on the gate as a function of ρ , g , R , and h . [5]

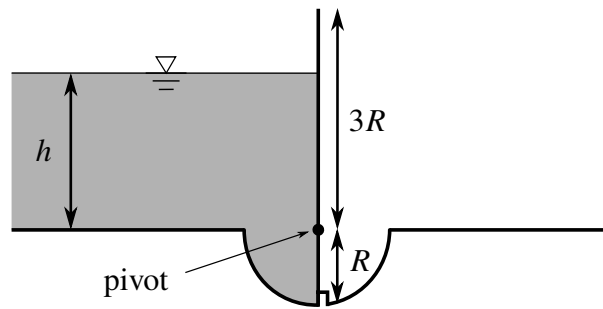


Fig. 1